Credit Name: CSE 2110 Procedural Programming 1

Assignment Name: AddCoins.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

I looked at how coin values work mathematically and figured out I needed to store the counts as integers and multiply them by their decimal values to get the correct total.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Combining different data types (int for coin counts and double for the math) was a bit tricky at first, but I realized Java automatically handles widening conversion when you multiply an int by a double.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes, I broke it down into getting user input for all four coins one by one, writing the calculation expression, and then printing the output string.

How did you decide on the tools, data structures, or algorithms to use?

I used Scanner for getting keyboard input, int variables for the coin amounts since you can't have a fraction of a coin, and a double for the total since money involves decimals.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I wrote it in pieces: first setting up the Scanner and prompts for quarters and dimes, then adding nickels and pennies, and finally writing the math formula and print statement.

How did you test your solution along the way to make sure it was working?

I ran test cases with easy numbers in my head (like 4 quarters to make $1.00) to make sure the math equation was calculating correctly before moving on

Overcoming Challenges

What part of the problem was the most difficult for you?

Making sure the decimal math worked properly and avoiding syntax errors with the parentheses in the calculation line.

How did you handle moments when you felt stuck or unsure of what to do next?

I checked back at my class notes on how to initialize the Scanner and took it line-by-press-line to trace where a semicolon or variable name might be missing.

Learning

Was there anything you learned that you think will help you with future challenges?

I got a lot more comfortable combining user inputs with arithmetic operations, which will definitely help for any future programs involving currency or calculations.